Deep Learning Model for Detection of Forest Fires

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Abstract

The frequency of forest fires in Alaska has increased fivefold over the past 15 years [1]. This project implements a range of deep-learning models for forest fire detection using a pre-compiled IEEE dataset [2]. The project aims to explore and research a variety of Neural Network architectures. We will discuss the architecture and modeling techniques and analyze the statistical results of these models to help provide a deeper understanding of neural network architecture in image classification.

This project will specifically explore the efficacy of three Convolutional Neural Network models (CNN) and ResNet. We will implement TensorFlow to build the models that will classify forest fires. The model will be trained to categorize images as 'Fire' or 'No Fire'. We will be analyzing the results of these models with the end goal being to provide an in-depth analysis of the model building process, accuracy, and loss of these models.

Deep learning, Fire Detection, Emergency Response, Image Classification, TensorFlow, Convolutional Neural Network, CNN, ResNET.

I. Introduction

Forest fires have devastating environmental and economic impacts, especially in regions like Alaska. Due to environmental conditions, these fires have significantly increased over the past two decades [6]. The primary motivation behind this project is to explore different model architectures, activation functions, and model accuracy rates while classifying thousands of images to determine whether there is a forest fire as a binary output.

With this project, we aim to mirror previous experiments done with deep learning models [3] [4] to explore potential forest fire detection solutions using a large normalized dataset of 254x254 RGB .jpeg format images provided by IEEE. The image dataset contains 39,375 training images and 8,617 testing images. The images consist of aerial imagery of Alaska forests with and without fires. The models will be trained on this dataset to be analyzed in this research.

After the images are analyzed, the binary output will be reviewed to provide a numerical value of percent accuracy and precision, measured by loss. The loss is computed using a cross-entropy loss function, which measures the divergence between the predicted probabilities and the true labels.

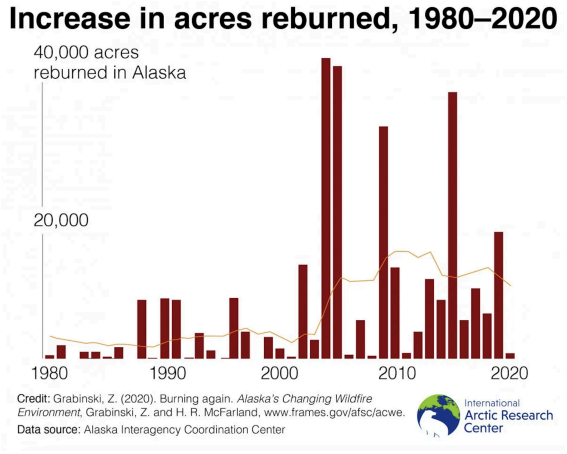


Fig. 1. Since 1980, more than 1,500 fires have occurred in areas that had previously burned within the past 20 years. The number of acres affected by these "reburn fires" has risen sharply each year since the early 2000s, as shown in this graph. [1]

II. Background

A. NumPy

NumPy is a fundamental library for numerical computing in Python. It supports large, multi-dimensional arrays and matrices and contains a collection of high-level mathematical functions to operate on these arrays. In this project, we will use NumPy for data manipulation, including handling image arrays and applying mathematical operations in the preprocessing and model development phases.

B. Pandas

Pandas is a powerful Python library for data manipulation and analysis. It provides data structures like DataFrames, allowing easy manipulation of labeled data. For this project, Pandas will be utilized to manage and preprocess metadata associated with the images and track and analyze results during the model evaluation phase.

A. MatplotLib

MatplotLib is a library used to create static, animated, and interactive data visualizations in Python. The selectable plots are line graphs, bar charts, histograms, and scatter plots, making it a versatile tool for data representation. Its simplicity, extensive customization options, and seamless integration with other Python libraries, notably Numpy and Pandas, are key advantages. MatplotLib will be utilized for this project to visualize training data, loss curves, and model performance.

B. Tensorflow

TensorFlow is an open-source machine learning framework designed for building and deploying deep learning models. It offers a flexible and scalable platform with tools for creating neural networks and processing large data sets efficiently. The main advantages of using TensorFlow include support for numerous neural networks, GPU acceleration, and a comprehensive system with libraries like Keras for easy model-building. This project will use TensorFlow to train convolutional neural networks to detect forest fires.

III. Data Description

The primary data source for this project is the IEEE Flame dataset, which consists of aerial imagery for fire detection. This dataset includes a total of about 40,000 254x254 frames of “Fire-vs-NoFire” training and testing data sets. The images vary by lighting conditions and vibrancy from image to image. Key model statistics such as accuracy and loss will be analyzed to ensure comprehensive model training and to provide a deeper understanding of model architecture techniques.

Our initial CNN model includes the layers outlined in figure 1. From this baseline, we have structured additional models to reduce loss and increase accuracy.

1. **Data Collection and Preprocessing:** RGB images will be utilized and obtained through the IEEE Flame dataset.
2. **Model Training and Compression:** A baseline CNN is initially used, then adapted to attempt to achieve better accuracy. An additional pre-trained model is used. The second and third CNN models follow a similar structure.

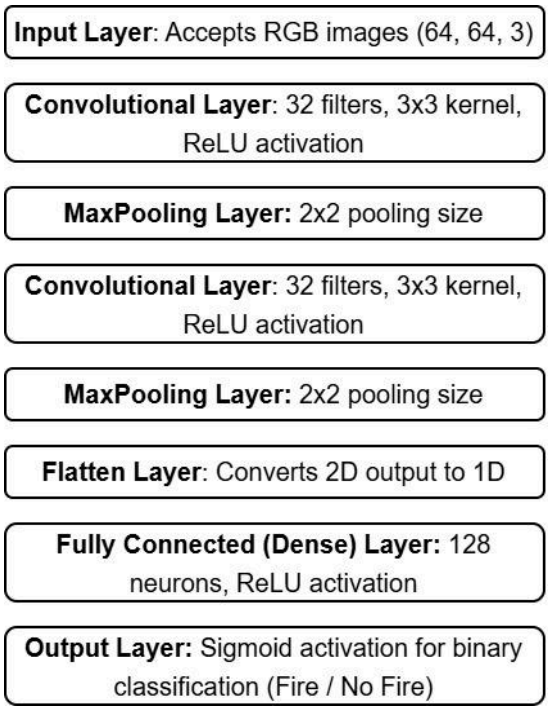


Fig. 2. Model 1 Architecture Design

IV. Methodology

To enhance model robustness and prevent overfitting, data augmentation was applied using TensorFlow's ImageDataGenerator. Techniques such as rescaling, rotation, width/height shifts, shear transformations, zooming, and horizontal flipping were used to introduce variations in the dataset, ensuring the model could generalize to unseen data. Images were resized to match model requirements, with Models 1 and 2 using 254x254, while Model 3 used 224x224 to align with ResNet50's pretrained architecture. All models were trained and evaluated on binary class data (fire/no fire) with accuracy and loss metrics.

Model 1 was designed as a baseline convolutional neural network (CNN) with a straightforward architecture featuring two convolutional layers (32 and 64 filters, 3x3 kernel size) with ReLU activation and max-pooling layers to downsample the feature maps. These were followed by a fully connected dense layer (128 neurons) with dropout (50%) for regularization and a sigmoid-activated output layer for binary classification.

Model 2 was designed to address the overfitting issues observed in Model 1 by introducing regularization and class-weight adjustments. Reducing the convolutional filters to 16 and 32 for the first and second layers, respectively, and applied L2 regularization (penalty term) to both convolutional and dense layers. Dropout was increased to 60% to further improve generalization.

Model 3 utilized a transfer learning approach with ResNet50, a state-of-the-art pretrained CNN. ResNet50 (with ImageNet weights) served as the base model, excluding its top classification layers. A GlobalAveragePooling2D layer reduced dimensionality, followed by a fully connected dense layer (128 neurons, ReLU activation) with 50% dropout for regularization and a sigmoid output layer for binary classification.

Areas for future exploration include further optimizing class balance, fine-tuning the pretrained layers in Model 3, and experimenting with additional architectures such as MobileNet or EfficientNet for a balance of speed and performance.

A. Software Architecture

The software architecture for this project was crafted to enable the efficient development, training, and evaluation of deep-learning models for forest fire detection. Leveraging Python and powerful libraries such as TensorFlow, NumPy, Pandas, and Matplotlib, the design ensured scalability, modularity, and robustness. The primary goal was to explore various neural network architectures and assess their suitability for accurately detecting fire in aerial imagery. The key components of the architecture included data preprocessing, model definition, training routines, and evaluation frameworks. Three distinct models were implemented to experiment with architectural variations:

Model 1 featured a simple architecture with two convolutional layers (32 and 64 filters) followed by max-pooling and a dense layer (128 neurons) with dropout (50%) for regularization. It was trained using the Adam optimizer and binary cross-entropy loss. While Model 1 achieved high training accuracy, its significant gap in validation accuracy highlighted overfitting.

Model 2 addressed overfitting by incorporating regularization techniques, including L2 regularization on convolutional and dense layers, and increasing dropout to 60%. Convolutional filters were reduced to 16 and 32 to minimize overparameterization. Class imbalance was handled using class weights, and callbacks like EarlyStopping and ReduceLROnPlateau were employed to monitor validation performance and adjust learning rates dynamically. These refinements resulted in improved test accuracy and reduced loss compared to Model 1.

Model 3 utilized transfer learning with the pretrained ResNet50 architecture. The ResNet50 base layers were frozen to retain pre-trained features, and a GlobalAveragePooling2D layer, a dense layer (128 neurons, 50% dropout), and a sigmoid output layer were added. This approach minimized training time for the base layers while focusing on fine-tuning the custom top layers. Despite its complexity, Model 3 achieved competitive performance, demonstrating the effectiveness of transfer learning for fire detection.

B. Analysis and Modeling Technique

The technique employed in this study centered on CNNs, which are well-suited for image classification tasks due to their ability to capture spatial patterns in visual data. All three models utilized a binary output with sigmoid activation to classify images as "Fire" or "No Fire," and training was conducted using the Adam optimizer with binary cross-entropy as the loss function.

To address overfitting, Model 2 introduced L2 regularization in convolutional and dense layers and increased dropout to 60%. Class weights were used to account for the imbalanced dataset, prioritizing fire detection accuracy. For Model 3, transfer learning with ResNet50 allowed leveraging pre-trained weights, drastically reducing training time while maintaining competitive accuracy. This iterative modeling technique, combined with callbacks such as EarlyStopping and learning rate scheduling, provided the flexibility to refine architectures and improve generalization. These design choices ensured the models were efficient and effective in detecting fire in diverse and challenging conditions.

To further enhance model performance, the study emphasized the importance of data augmentation and preprocessing techniques. Augmentation methods, including rotation, zooming, horizontal flipping, and scaling, were applied during training to diversify the dataset and improve the model's ability to generalize across unseen data. Rescaling the pixel values to a range of [0, 1] ensured that all input features had consistent scales, facilitating more stable and efficient training. Additionally, transfer learning in Model 3 leveraged the pretrained ResNet50 architecture, enabling the model to extract robust high-level features from images while requiring fewer computational resources for training. This approach demonstrated the potential of pretrained models in handling complex classification tasks with limited custom training. By combining advanced architectures, thoughtful regularization techniques, and efficient data handling, the methodology ensured a systematic exploration of CNN capabilities for forest fire detection.

V. Results and Discussion

The results of this study illustrate the strengths and limitations of using CNNs for forest fire detection, particularly regarding accuracy, loss, and computational efficiency.

Model 1, a baseline architecture with two convolutional layers, achieved a high training accuracy of approximately 94.4%. However, its validation accuracy dropped sharply to 48.4%, with a test loss of 2.8972. This significant gap between training and validation performance suggested overfitting, where the model captured noise in the training data but failed to generalize to unseen images. The model required 27 minutes and 31 seconds to compile, highlighting its relatively lightweight design despite its limitations in performance.

Model 2 introduced regularization techniques, class weights, and adjustments to dropout rates to address the overfitting observed in Model 1. These refinements resulted in improved validation accuracy of 58.6% and a reduced test loss of 0.8966, signifying better generalization. The model was also more computationally efficient, with a compile time of 23 minutes and 31 seconds. These enhancements demonstrated the importance of balancing architectural complexity and regularization to achieve more robust performance in challenging datasets with class imbalances.

Model 3 utilized transfer learning with ResNet50, leveraging pre-trained weights to extract high-level features. This model achieved a validation accuracy of 58.3% and a test loss of 0.8996. While its compile time was longer at 38 minutes and 27 seconds due to the complexity of the ResNet50 architecture, it demonstrated competitive performance, underscoring the potential of transfer learning for fire detection.

Overall, these results highlight the iterative process of refining model architecture and training strategies to balance accuracy, loss, and computational efficiency.

Future work could focus on enhancing data quality, addressing class imbalances, and fine-tuning transfer learning models for even better performance.

VI. Acknowledgments and References

We would like to acknowledge the IEEE DataPort for providing access to the Flame dataset, which was integral to this project's development.

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